Software Requirements Specification (SRS)

Web-Based Laboratory Inventory and Resource Management System

Scope: Junior High School Science Laboratory and Technology and Livelihood Education (TLE) Laboratory

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1. Introduction

1.1 Purpose

The Web-Based Laboratory Inventory and Resource Management System is a centralized web application for managing the Junior High School's laboratory resources, equipment, supplies, borrowing transactions, maintenance records, inventory movements, and user accountability.

The initial implementation shall support the school's Science Laboratory and Technology and Livelihood Education (TLE) Laboratory.

The system is intended to replace or reduce the use of paper-based inventory records, disconnected spreadsheets, and manually maintained borrowing records.

The primary purpose of the system is to provide authorized personnel with an accurate and centralized way to determine what resources the school has, where they are located, what condition they are in, how many are available, who is currently using them, and what has happened to them throughout their lifecycle.

The system shall prioritize inventory accuracy, accountability, security, usability, and maintainability.

2. Scope

2.1 In-Scope Features

The system shall provide the following capabilities:

1. User authentication
2. Role-based authorization
3. User account management
4. Laboratory management
5. Inventory management
6. Inventory categorization
7. Consumable inventory tracking
8. Individually tracked equipment and assets
9. Inventory stock-in and stock-out
10. Inventory adjustments
11. Equipment condition monitoring
12. Equipment status monitoring
13. Borrowing requests
14. Borrowing approval
15. Item release
16. Item return
17. Damaged and lost item recording
18. Equipment maintenance records
19. Inventory search and filtering
20. Inventory history
21. Dashboard
22. Reports
23. Audit logging
24. Data validation
25. Basic backup and recovery mechanisms

2.2 Out-of-Scope Features

The following features are not required for the initial version:

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26. Artificial intelligence
27. Machine learning
28. AI chatbot
29. Demand forecasting
30. Predictive analytics
31. AI recommendations
32. Computer vision
33. Facial recognition
34. RFID
35. Barcode or QR scanning
36. Mobile application
37. Automated purchasing
38. Online payment
39. Supplier marketplace
40. IoT integration
41. Automated procurement
42. Predictive maintenance

These features may be considered future enhancements after the core system has been successfully implemented and evaluated.

3. System Objectives

O1. Centralize laboratory information. Provide a single database for laboratory inventory and resource information.

O2. Improve inventory accuracy. Provide controlled mechanisms for recording inventory additions, removals, adjustments, borrowing, returns, damage, loss, and disposal.

O3. Improve accountability. Maintain records identifying users responsible for significant inventory-related transactions.

O4. Monitor resource condition. Allow authorized personnel to record and monitor the condition and availability of laboratory equipment.

O5. Improve borrowing management. Provide a structured process for requesting, approving, releasing, and returning laboratory resources.

O6. Reduce manual record keeping. Reduce dependence on paper forms and manually maintained spreadsheets.

O7. Improve information accessibility. Allow authorized users to quickly search and retrieve laboratory resource information.

O8. Support administrative reporting. Provide reports and summaries regarding laboratory resources, transactions, equipment, and usage.

O9. Provide a secure foundation for future enhancements. Maintain an architecture capable of supporting future intelligent features without making those features dependencies of the core inventory system.

4. Stakeholders

43. Junior High School Administration: Requires overall resource visibility, accountability, and reporting.
44. Science Laboratory Coordinator: Manages Science Laboratory resources.
45. TLE Laboratory Coordinator: Manages TLE Laboratory resources.
46. Teachers: View resources and request laboratory items.
47. System Administrator: Manages users, security, and system configuration.
48. Students: Indirect beneficiaries of properly managed laboratory resources.

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5. User Roles

5.1 System Administrator

The System Administrator shall be responsible for system-wide administration.

Permissions:

49. Create user accounts
50. Modify user accounts
51. Activate or deactivate accounts
52. Assign user roles
53. Assign laboratory access
54. Manage laboratory records
55. Manage system categories
56. View system-wide inventory
57. View system-wide reports
58. View audit logs
59. Manage system configuration

5.2 Laboratory Coordinator

A Laboratory Coordinator shall be responsible for one or more assigned laboratories. Examples include the Science Laboratory Coordinator and TLE Laboratory Coordinator.

Permissions:

60. View assigned laboratory inventory
61. Add inventory items
62. Modify inventory records
63. Register equipment
64. Manage equipment status
65. Manage equipment condition
66. Approve borrowing requests
67. Process returns
68. Record damage and loss
69. Record maintenance
70. Perform inventory adjustments
71. View laboratory history
72. Generate laboratory reports

5.3 Teacher

Teachers shall primarily use the system to check resource availability and manage borrowing requests.

Permissions:

73. View available inventory
74. Search inventory
75. Submit borrowing requests
76. View their own requests
77. View request status
78. View their own borrowing history
79. Confirm or facilitate returns

Teachers shall not directly modify inventory quantities.

5.4 Student

- Students shall not require individual system accounts in Version 1.0.
- Students shall interact with laboratory resources through their teachers and laboratory personnel.

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5.5 Laboratory Staff

Laboratory Staff shall perform authorized day-to-day laboratory operations. The role may be assigned to the Laboratory Coordinator or to separate staff members, depending on the school's operational arrangement.

Authorized day-to-day operations may include item release, return processing, damage and loss recording, maintenance recording, and other laboratory operations permitted by the user's assigned permissions.

6. Laboratory Management

6.1 Laboratory Information

The system shall treat a laboratory as a configurable entity rather than hard-coding Science and TLE laboratories into the application.

Each laboratory shall have:

80. Laboratory ID
81. Laboratory name
82. Laboratory type
83. Location
84. Description
85. Status
86. Date created

Example:

- LAB-001: Science Laboratory, Type: Science
- LAB-002: TLE Laboratory, Type: TLE

The architecture shall permit additional laboratories to be added in the future without significant changes to the system.

7. Inventory Classification

7.1 Consumable Inventory

Consumable inventory shall primarily be tracked by quantity.

Examples:

87. Filter paper
88. Chemicals
89. Thread
90. Fabric
91. Food ingredients
92. Glue
93. Batteries
94. Electrical components

Example:

- Item: Filter Paper
- Unit: Sheet
- Quantity: 250

7.2 Equipment and Assets

Equipment that requires individual accountability shall be tracked separately.

Examples:

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95. Microscope
96. Sewing machine
97. Drill
98. Weighing scale
99. Computer
100. Soldering iron

Example:

- Asset ID: MIC-003
- Item: Microscope
- Laboratory: Science Laboratory
- Status: Available
- Condition: Good

8. Functional Requirements

FR-001. User Authentication. The system shall allow authorized users to authenticate using valid credentials. The system shall accept valid login credentials, reject invalid credentials, establish an authenticated session, provide logout functionality, and prevent unauthorized access to protected resources. Passwords shall never be stored in plain text.

FR-002. User Management. Authorized administrators shall be able to create and update users, assign roles and laboratory permissions, activate and deactivate accounts, and reset user credentials when authorized.

FR-003. Role-Based Authorization. The system shall enforce access restrictions based on user roles. Authorization shall be enforced on the server side. Hiding a button or page element shall not be considered sufficient authorization.

9. Inventory Management Requirements

FR-004. Register Inventory Item. Authorized users shall be able to register inventory items. An inventory record may contain Item ID, Item name, Description, Category, Laboratory, Item type, Unit of measurement, Quantity, Minimum stock level, Condition, Location, Acquisition date, Acquisition source, Remarks, and Status.

FR-005. Modify Inventory Item. Authorized users shall be able to update inventory information. Changes to significant inventory information shall be recorded in the audit log.

FR-006. Archive Inventory Item. The system shall provide an archive mechanism instead of permanently deleting inventory records with transaction history. Archived items shall remain available for historical reporting.

10. Category Management

FR-007. Manage Inventory Categories. The system shall support configurable inventory categories. Example Science categories include Laboratory Equipment, Glassware, Chemicals, Measuring Instruments, Specimens, Safety Equipment, and Consumables. Example TLE categories include Hand Tools, Power Tools, Sewing Equipment, Cooking Equipment, Electrical Components, Raw Materials, Consumables, and Safety Equipment. Categories shall not be permanently limited to these examples.

11. Quantity Management

FR-008. Stock-In. Authorized users shall be able to record incoming inventory. The transaction shall include Item, Quantity received, Date, User, Source or reason, and Remarks.

FR-009. Stock-Out. Authorized users shall be able to record inventory leaving stock. Possible reasons include Consumption, Damage, Loss, Disposal, Transfer, and Other authorized reasons.

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FR-010. Inventory Adjustment. Authorized users shall be able to make controlled inventory adjustments. An adjustment shall require a reason. Example: System Quantity 50, Physical Count 47, Adjustment -3, Reason Physical inventory discrepancy.

12. Equipment and Asset Management

FR-011. Register Asset. The system shall allow authorized personnel to register individually tracked assets. Each asset shall have a unique Asset ID.

FR-012. Asset Status. The system shall support the statuses Available, In Use, Borrowed, Under Inspection, Under Maintenance, Damaged, Lost, and Retired.

FR-013. Asset Condition. The system shall maintain condition separately from operational status. Possible conditions are Excellent, Good, Fair, Damaged, and Non-functional.

13. Borrowing Management

FR-014. Create Borrowing Request. Authorized teachers and staff shall be able to create borrowing requests. A request shall contain Requester, Laboratory, Purpose, Request date, Expected return date, Requested items, Requested quantities, and Remarks.

FR-015. Borrowing Request Status. The system shall support the statuses Pending, Approved, Rejected, Released, Returned, Overdue, and Cancelled.

FR-016. Approve or Reject Request. Authorized Laboratory Coordinators or Staff shall be able to approve or reject requests. Rejected requests shall contain a reason where appropriate.

FR-017. Release Items. When items are physically released, authorized staff shall record the release. The system shall update the relevant inventory or asset status.

FR-018. Return Items. Authorized staff shall be able to process returned items. The return record shall include Return date, Quantity returned, Condition, Missing quantity, Damaged quantity, and Remarks.

14. Damage and Loss Management

FR-019. Record Damaged Item. Authorized users shall be able to record damaged resources. The record shall include Item or Asset, Date, Damage description, Reported by, Current condition, Action taken, and Remarks.

FR-020. Record Lost Item. Authorized personnel shall be able to mark an item or asset as lost. A lost item shall not be considered available.

15. Maintenance Management

FR-021. Create Maintenance Record. Authorized users shall be able to record equipment maintenance. A maintenance record shall include Asset, Reported issue, Date reported, Maintenance action, Person responsible, Date completed, Maintenance status, and Remarks.

FR-022. Maintenance History. Each individually tracked asset shall maintain a historical record of maintenance activities.

16. Search and Filtering

FR-023. Search Inventory. Users shall be able to search inventory using Item name, Item ID, Asset ID, Category, Laboratory, Location, Status, and Condition.

FR-024. Filter Inventory. Users shall be able to filter inventory by Laboratory, Category, Item type, Availability, Condition, Status, and Location.

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FR-025. Sort Inventory. Users shall be able to sort inventory by Name, Quantity, Date added, Condition, and Status.

17. Dashboard

FR-026. Role-Based Dashboard. The system shall provide a dashboard appropriate to the user's role. A Laboratory Coordinator may see laboratory-level summaries such as Total Inventory, Available Equipment, Low Stock Items, Under Maintenance, Pending Requests, and Overdue Returns. A System Administrator may see an overall school-level summary. The dashboard shall not depend on AI functionality.

Illustrative dashboard values from the original specification are retained as examples only and are not system requirements:

- Science Laboratory
- Total Inventory: 482
- Available Equipment: 91
- Low Stock Items: 12
- Under Maintenance: 5
- Pending Requests: 4
- Overdue Returns: 2

18. Reporting Requirements

FR-027. Inventory Report. The report shall include Item, Category, Quantity, Laboratory, Condition, Status, and Location.

FR-028. Borrowing Report. The report shall include Borrower, Item, Quantity, Date borrowed, Expected return, Actual return, and Status.

FR-029. Equipment Report. The report shall include Asset ID, Equipment, Laboratory, Condition, Status, and Maintenance status.

FR-030. Inventory Movement Report. The report shall include Stock-in, Stock-out, Adjustments, Transfers, Date, User, and Reason.

FR-031. Report Output. Reports shall be printable and may support PDF or spreadsheet export.

19. Audit Logging

FR-032. Record User Activities. The system shall maintain an audit trail for significant system activities. Examples include Login, Logout, Inventory creation, Inventory modification, Inventory adjustment, Borrowing approval, Item release, Item return, Asset status change, Maintenance record creation, and User modification.

An audit record shall contain, where applicable:

101. Timestamp
102. User
103. Action
104. Affected Entity
105. Record ID
106. Previous Value
107. New Value
108. Reason

Audit records shall not be editable by ordinary users.

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20. Data Validation Requirements

FR-033. Validate Stored Data. The system shall validate data before storing it.

Validation shall include:

109. Quantity cannot be negative.
110. Required fields cannot be empty.
111. Asset IDs must be unique.
112. Inventory IDs must be unique.
113. Return quantity cannot exceed released quantity.
114. Unavailable assets cannot be released.
115. Lost assets cannot be borrowed.
116. Items under maintenance cannot be released.
117. Deactivated users cannot create new borrowing requests.
118. Inventory adjustments must contain a valid reason.

21. Business Rules

BR-001. Business Rule. Every inventory item shall belong to a laboratory unless designated as shared inventory.

BR-002. Business Rule. Every individually tracked asset shall have a unique Asset ID.

BR-003. Business Rule. Inventory quantities shall never be negative.

BR-004. Business Rule. An item marked as Lost shall not be considered available.

BR-005. Business Rule. An asset under maintenance shall not be available for borrowing.

BR-006. Business Rule. A borrowing request must be approved before the items can be released.

BR-007. Business Rule. Only authorized users may modify inventory quantities.

BR-008. Business Rule. Completed transactions shall not be silently deleted.

BR-009. Business Rule. Important inventory changes shall create audit records.

BR-010. Business Rule. Archived inventory shall remain available for historical reports.

BR-011. Business Rule. A Laboratory Coordinator shall only manage laboratories assigned to them unless granted system-wide privileges.

BR-012. Business Rule. A Teacher shall only be able to view and manage their own borrowing requests unless additional permissions are granted.

22. Security Requirements

SR-001. Authentication Security. The system shall use secure password hashing. It shall never store plain text passwords, shall protect authenticated sessions, shall provide secure logout, and shall prevent unauthorized session access.

SR-002. Authorization Security. All protected operations shall verify the authenticated user's permissions on the server.

SR-003. Input Security. The system shall validate and sanitize user-provided data. It shall implement protection against common web application vulnerabilities, including SQL injection, cross-site scripting (XSS), cross-site request forgery (CSRF), broken access control, and session-related attacks.

SR-004. HTTPS. Production deployment shall use HTTPS to protect data transmitted between the browser and server.

SR-005. Auditability. Security-sensitive and inventory-sensitive operations shall be logged.

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23. Data Integrity Requirements

DIR-001. Maintain Inventory Consistency. The system shall maintain consistency between inventory records and transactions.

The system shall prevent normal application operations from producing inconsistent states. For example, a state in which Available is 5 and Borrowed is 7 shall not be permitted when the underlying inventory does not support it.

DIR-002. Atomic Related Updates. Database transactions should be used where multiple related records must be updated together.

24. Performance Requirements

PR-001. Normal Response Time. The system shall provide acceptable response times for expected school usage. Normal operations such as login, inventory search, viewing inventory, creating requests, processing returns, and updating inventory should respond without unnecessary delay under normal operating conditions.

PR-002. Expected Usage. The system is not required to support internet-scale traffic. It shall instead be optimized for the expected number of school users and laboratory records.

25. Usability Requirements

UR-001. Simple and Consistent Interface. The interface shall be simple and consistent and shall use understandable terminology.

UR-002. Clear Validation. The system shall provide clear validation messages.

UR-003. Resource Availability. The interface shall clearly distinguish available and unavailable resources.

UR-004. Supported Screen Sizes. The system shall work on desktop and tablet-sized screens.

UR-005. Efficient Common Operations. The interface shall minimize unnecessary steps for common operations.

UR-006. Confirmation. The system shall provide confirmation for destructive or significant actions.

UR-007. Technical Knowledge. The system should be usable by personnel without advanced technical knowledge.

26. Maintainability Requirements

MR-001. Modular Architecture. The application shall use a modular architecture.

Logical modules should include:

119. Authentication
120. Users
121. Laboratories
122. Categories
123. Inventory
124. Assets
125. Borrowing
126. Maintenance
127. Reports
128. Audit

MR-002. Separation of Responsibilities. The system should separate the Presentation, Business Logic, Data Access, and Database layers. This separation will make the application easier to maintain and expand.

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27. Scalability Requirements

SC-001. Additional Laboratories. Although the initial implementation covers Science and TLE, the system architecture shall allow additional laboratories to be added. Examples include Science Laboratory, TLE Laboratory, Agriculture Laboratory, and Industrial Arts Laboratory.

SC-002. Configuration-Based Expansion. Adding a laboratory should primarily require configuration or data entry rather than changes to the fundamental system architecture.

28. Backup and Recovery

BRR-001. Database Backup Strategy. The system shall implement a database backup strategy.

The backup process should provide:

129. Regular database backups
130. Backup retention
131. Secure backup storage
132. Recovery procedures
133. Periodic verification of backups

The goal is to prevent permanent loss of inventory and transaction history.

29. System Principles

Principle 1. Accuracy. The database should represent the actual state of laboratory resources as accurately as possible.

Principle 2. Accountability. Significant changes must be attributable to an authorized user.

Principle 3. Security. Users should only access the information and functionality required by their role.

Principle 4. Traceability. Important resources and transactions should have a history.

Principle 5. Simplicity. The system should solve the laboratory's actual problems without unnecessary complexity.

Principle 6. Extensibility. The architecture should support additional laboratories and future capabilities.

Principle 7. Separation of Concerns. The presentation layer, business logic, database, authentication, and auditing should have clear responsibilities.

30. Definition of Success

DS-001. Core Workflow. The fundamental system shall be considered successful when authorized laboratory personnel can reliably perform the following workflow: register a laboratory resource, locate it, determine its quantity, status, and condition, allow an authorized user to request it, approve and release it, record its use, return it, record its condition, perform maintenance if necessary, and preserve the complete transaction history.

The system should allow a Laboratory Coordinator to determine, at any given time:

134. What do we have?
135. How many do we have?
136. Where is it?
137. What condition is it in?
138. Is it available?
139. Who currently has it?
140. When was it borrowed?
141. When should it be returned?
142. Has it been damaged or maintained?

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143. What happened to it historically?

31. Future Enhancement Boundary

FE-001. Future Intelligent Capabilities. The architecture shall permit future implementation of intelligent capabilities without making them fundamental dependencies of the inventory system.

Potential future extensions include:

144. Demand forecasting
145. Laboratory readiness analysis
146. Intelligent recommendations
147. Predictive maintenance
148. Additional analytics

The core system must remain fully functional without these future capabilities.

32. Final System Definition

The Web-Based Laboratory Inventory and Resource Management System is a secure, centralized, and auditable information system designed to manage the school's Science and TLE laboratory resources throughout their lifecycle, including registration, inventory tracking, borrowing, returning, condition monitoring, maintenance, and reporting.

Approval

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